

Partial Result Packet

Endpoint-positive tree products for the distortion 2 Hausdorff-Dvoretzky question

Status: partial result, likely valid

Source problem

The source paper is:

arXiv:1106.0879, Manor Mendel and Assaf Naor,

Ultrametric subsets with large Hausdorff dimension.

It appeared in *Inventiones Mathematicae* 192 (2013), 1–54.

The local copy is included as `source_paper.pdf`.

Question 1.3 of the source asks, for $\alpha > 0$ and $D > 1$, how large a Hausdorff-dimensional subset must exist in every compact metric space of Hausdorff dimension at least α if the subset is required to embed into Hilbert space with distortion D . In the low-distortion regime the source proves positive dimension at distortion $2 + \delta$, and proves that no positive-dimensional theorem is possible for distortions strictly smaller than 2.

Theorem 1.8. *For every $\alpha \in (0, \infty)$ there exists a compact metric space (X, d) of Hausdorff dimension α , such that if $S \subseteq X$ embeds into Hilbert space with distortion strictly smaller than 2 then $\dim_H(S) = 0$.*

It was recently observed in [14] that Theorem 1.8 easily implies the following seemingly stronger assertion: there exists a compact metric space X_∞ such that $\dim_H(X_\infty) = \infty$, yet every subset $S \subseteq X_\infty$ that embeds into Hilbert space with distortion strictly smaller than 2 must have $\dim_H(S) = 0$.

As in the case of the finite nonlinear Dvoretzky theorem, Question 1.3 at distortion 2 remains open. In the same vein, the correct asymptotic dependence on δ in (4) is unknown.

Theorem 1.7 follows from the following result in the spirit of Theorem 1.5, via the same Frostman measure argument.

Theorem 1.9. *There exists a universal constant $c \in (0, \infty)$ such that for every $\delta \in (0, 1/2)$ there exists $c'_\delta \in (0, \infty)$ with the following property. Every metric measure space (X, d, μ) has a closed subset $S \subseteq X$ such that (S, d) embeds into an ultrametric space with distortion $2 + \delta$, and for every $\{x_i\}_{i \in I} \subseteq X$ and $\{r_i\}_{i \in I} \subseteq [0, \infty)$ such that the balls $\{B(x_i, r_i)\}_{i \in I}$ cover S , we have*

The passage above, from page 6 of the source PDF, states that Question 1.3 at distortion 2 remains open.

Result proved here

This packet proves that the source's own $G(n, 1/2)$ -fractal lower-bound examples are not counterexamples at the endpoint. In fact, those compact spaces themselves embed into ultrametric spaces

with distortion 2. Combined with the source's lower bound for distortions strictly smaller than 2, this shows that the random-graph fractals are exactly sharp at the threshold.

Idea of the Proof

The source's lower-bound compacta are boundaries of trees whose branching levels carry finite metrics. Two boundary points are separated at their first different coordinate; their distance is the corresponding finite-level distance, multiplied by the level scale. If we replace each finite-level metric by the constant metric equal to its diameter, the resulting boundary distance is an ultrametric. Therefore the only possible distortion loss is the aspect ratio of the finite metrics used at the levels. The $G(n, 1/2)$ blocks have only two nonzero distances, 1 and 2, hence aspect ratio 2.

Theorem

Theorem. *Let $\alpha > 0$, let $n_0 = 1$ and $n_k > 1$, and for each $k \geq 1$ let d_k be a metric on $\{1, \dots, n_k\}$ normalized so that*

$$\text{diam}_{d_k}\{1, \dots, n_k\} = 1, \quad \min_{i \neq j} d_k(i, j) \geq \eta$$

for some fixed $\eta \in (0, 1]$. On

$$Y = \prod_{k=1}^{\infty} \{1, \dots, n_k\}$$

define, as in Section 10.1 of the source,

$$\rho_\alpha(x, y) = \frac{d_{k(x,y)}(x_{k(x,y)}, y_{k(x,y)})}{\prod_{i=0}^{k(x,y)-1} n_i^{1/\alpha}},$$

where $k(x, y)$ is the first coordinate at which x and y differ. Then (Y, ρ_α) embeds into an ultrametric space with distortion at most $1/\eta$.

Proof. Define

$$u_\alpha(x, y) = \begin{cases} 0, & x = y, \\ \left(\prod_{i=0}^{k(x,y)-1} n_i^{1/\alpha} \right)^{-1}, & x \neq y. \end{cases}$$

This is the usual boundary ultrametric of the rooted tree: for any three boundary points, the two largest among the three first-splitting levels coincide, so $u_\alpha(x, z) \leq \max\{u_\alpha(x, y), u_\alpha(y, z)\}$.

By the normalization of d_k , whenever $x \neq y$,

$$\eta u_\alpha(x, y) \leq \rho_\alpha(x, y) \leq u_\alpha(x, y).$$

Equivalently,

$$\rho_\alpha(x, y) \leq u_\alpha(x, y) \leq \eta^{-1} \rho_\alpha(x, y).$$

Thus the identity map from (Y, ρ_α) to (Y, u_α) is an embedding into an ultrametric space with distortion at most $1/\eta$. $\square$

Corollary. *For the $G(n, 1/2)$ -fractal examples used in Section 10.3 of the source to prove Theorem 1.8, the whole compact space embeds into an ultrametric space, and hence into Hilbert space, with distortion at most 2.*

Proof. In Section 10.3, the source uses finite random-graph metrics W_n in which the positive distances are 1 and 2. After normalizing the diameter to one, the nonzero distances are $1/2$ and 1, so $\eta = 1/2$ in the theorem. Therefore the resulting tree-product compactum is 2-equivalent to an ultrametric.

The source’s Lemma 10.2 and Theorem 10.5 compute the Hausdorff dimension of these tree products, and the proof of Theorem 1.8 then shows that every positive-dimensional subset of the $G(n, 1/2)$ -fractal incurs Hilbert distortion at least 2. Since the whole space embeds into an ultrametric with distortion at most 2, every subset embeds into Hilbert space with distortion at most 2. Thus, for this family, the endpoint threshold is attained exactly. $\square$

Verification and limitations

This is a partial result. It does not prove that every compact metric space of positive Hausdorff dimension has a positive-dimensional subset embedding into Hilbert space with distortion 2. It proves that one central attempted counterexample route, namely the source’s $G(n, 1/2)$ -fractal construction, cannot disprove the endpoint statement.

The proof uses only the explicit tree-product metric from Section 10.1 of the source and the two-valued nature of the $G(n, 1/2)$ metrics described in Section 10.3. The bounded literature check found the later Mendel–Naor paper *Ultrametric skeletons* (arXiv:1112.3416), which strengthens the source’s measure skeleton theorem and derives Talagrand’s nonlinear Dvoretzky theorem, but did not find a later full solution of the distortion 2 Hausdorff-dimensional endpoint.

References

- M. Mendel and A. Naor, *Ultrametric subsets with large Hausdorff dimension*, arXiv:1106.0879; Invent. Math. 192 (2013), 1–54.
- M. Mendel and A. Naor, *Ultrametric skeletons*, arXiv:1112.3416; Proc. Natl. Acad. Sci. USA 110 (2013), no. 48, 19256–19262.
- Y. Bartal, N. Linial, M. Mendel, and A. Naor, *On metric Ramsey-type phenomena*, Ann. of Math. 162 (2005), 643–709.